

CO2-AT2

MLA0201 - Fundamentals Of Machine Learning Scenario-Based

Q1. Linear Regression for House Price Estimation

Algorithm Used:

Linear Regression

Model / Formula:

House Price = $\square$ ■ + $\square$ ■(Area) + $\square$ ■(Bedrooms) + $\square$ ■(Location Score)

Features:

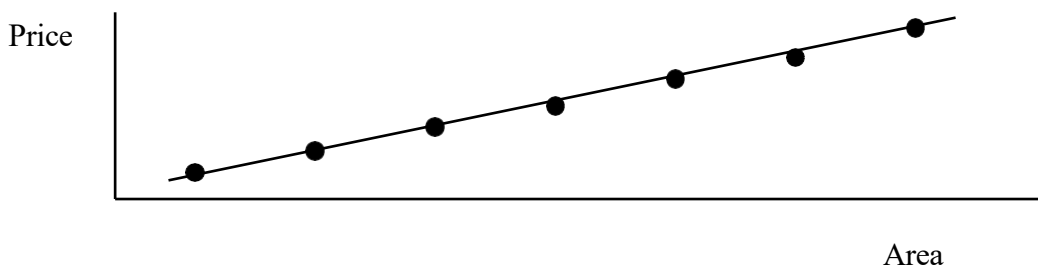
- Area = Built-up area in sq.ft
- Bedrooms = Number of bedrooms
- Location Score = 1 (Low), 2 (Medium), 3 (Prime)

Steps:

1. Collect historical house-price records.
2. Clean missing values and normalize numerical features.
3. Train the Linear Regression model.
4. Estimate the regression coefficients.
5. Use the trained model to estimate the price of a new house.

Related Visual:

House Area vs. Estimated Price



Expected Output:

Area = 1500 sq.ft

Bedrooms = 3

Location Score = 3 (Prime)

Predicted Price = ■72,00,000

Conclusion:

Linear Regression estimates house prices by learning the relationship between property features and a continuous target value.

Q2. Linear Classification for Email Spam Detection

Algorithm Used:

Linear Classification

Model / Formula:

$$f(x) = w_0 + w_1x_1 + w_2x_2$$

Features:

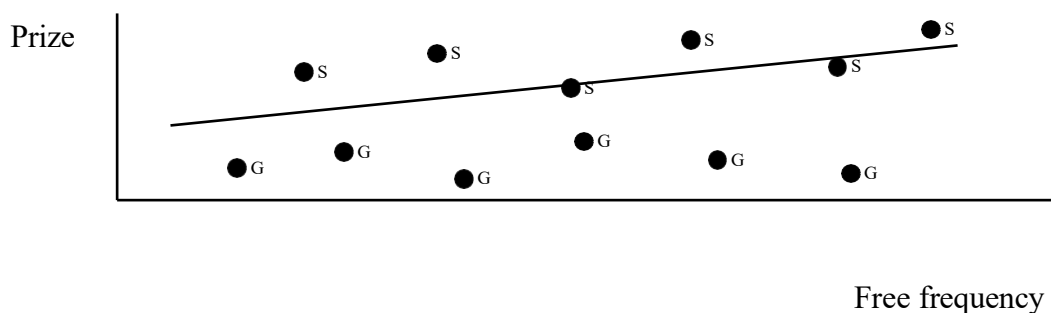
- x_1 = Frequency of the word "Free"
- x_2 = Frequency of the word "Prize"

Steps:

1. Collect labelled spam and genuine emails.
2. Convert messages into numerical word-frequency features.
3. Train the linear classifier.
4. Determine the separating decision boundary.
5. Classify a new email as spam or genuine.

Related Visual:

Spam / Genuine Email Feature Space



Expected Output:

Free = 5

Prize = 3

Conclusion:

Linear classification separates two email classes by using a decision boundary learned from labelled examples.

Q3. Naïve Bayes for Cold/Flu Prediction

Algorithm Used:

Naïve Bayes Classifier

Model / Formula:

$$P(\text{Class} | X) = P(\text{Class}) P(X | \text{Class}) / P(X)$$

Features:

- Fever
- Cough
- Headache

Steps:

1. Calculate the prior probability of each class.
2. Find the conditional probability of each symptom.
3. Apply Bayes' theorem to the observed symptoms.
4. Compare the posterior probabilities.
5. Select the class with the highest probability.

Related Visual:

Symptoms □ Posterior Probability



Expected Output:

Fever = Yes

Cough = Yes

Conclusion:

Naïve Bayes uses probability distributions of symptoms to make a fast classification decision.

Q4. Discriminant Function for Loan Approval

Algorithm Used:

Linear Discriminant Analysis (LDA)

Model / Formula:

$$g(x) = w_{\text{Intercept}} + w_{\text{Income}}(\text{Income}) + w_{\text{Credit Score}}(\text{Credit Score})$$

Features:

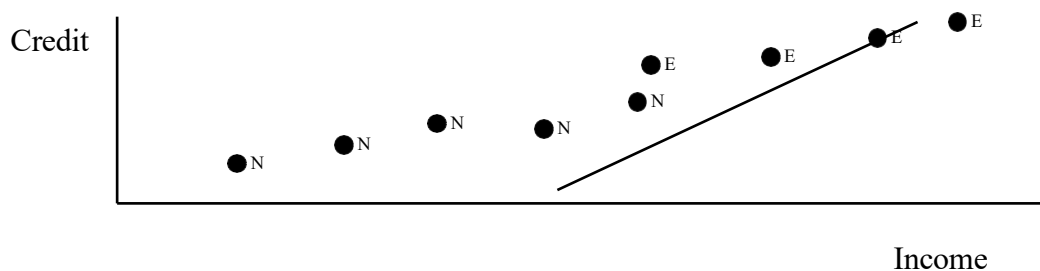
- Income = Monthly applicant income
- Credit Score = Numerical creditworthiness score

Steps:

1. Collect applicant records with approval labels.
2. Calculate the mean of each class.
3. Construct the discriminant function.
4. Determine the decision boundary.
5. Classify a new applicant as Eligible or Not Eligible.

Related Visual:

Income vs. Credit Score: LDA Boundary



Expected Output:

Income = 85,000

Credit Score = 790

Prediction = Eligible

Conclusion:

LDA separates loan applicants by finding a linear boundary between eligible and non-eligible groups.

Q5. Probabilistic Generative Model for Digit Recognition

Algorithm Used:

Gaussian Naïve Bayes

Model / Formula:

$$P(\text{Class} | \text{Image}) \propto P(\text{Class}) P(\text{Image} | \text{Class})$$

Features:

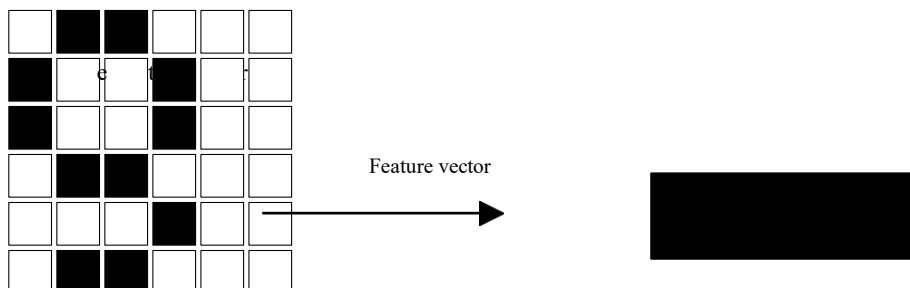
- Pixel intensity values
- Extracted image features
- Digit class probabilities

Steps:

1. Collect labelled handwritten digit images.
2. Convert each image into numerical features.
3. Estimate the distribution of features for each digit.
4. Calculate posterior probabilities for the input image.
5. Choose the digit with the highest posterior probability.

Related Visual:

Example



Expected Output:

Input = Handwritten digit image

Predicted Digit = 8

Conclusion:

Generative probability models recognize handwritten digits by modelling how feature values are distributed within each digit class.

Q6. Probabilistic Discriminative Model for Customer Churn

Algorithm Used:

Logistic Regression

Model / Formula:

$$P(Y=1) = 1 / (1 + e^{-z}), \text{ where } z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

Features:

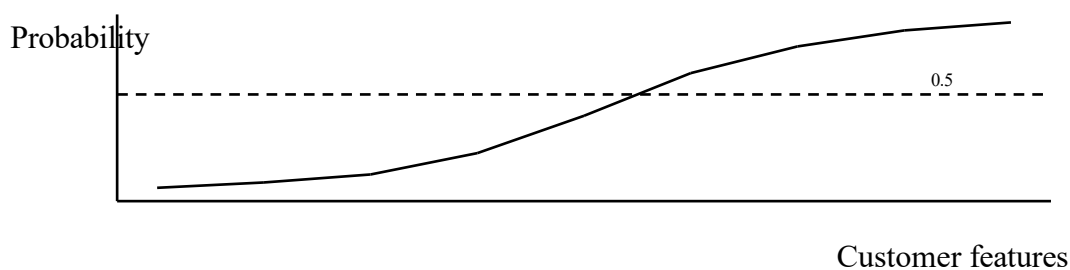
- Call Duration
- Internet Usage
- Monthly Bill

Steps:

1. Collect customer usage and billing records.
2. Train the logistic regression model.
3. Estimate the model coefficients.
4. Calculate the probability of churn.
5. Apply a threshold to decide whether the customer is likely to churn.

Related Visual:

Customer churn probability



Expected Output:

Call Duration = 210

min Internet Usage

= High Monthly Bill

= 1,499

Probability = 0.78

Prediction = Customer Will Churn

Conclusion:

Logistic Regression converts customer features into a probability and is useful for binary churn prediction.

Q7. Bayesian Logistic Regression for Online Purchase Prediction

Algorithm Used:

Bayesian Logistic Regression

Model / Formula:

$P(\text{Purchase}) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1(\text{Ad Views}) + \beta_2(\text{Clicks})$

Features:

- Advertisement views
- Number of ad clicks
- Prior information about customer behaviour

Steps:

1. Collect advertisement interaction data.
2. Choose prior distributions for model parameters.
3. Estimate posterior coefficients using the observed data.
4. Calculate purchase probability.
5. Predict whether the customer is likely to purchase.

Related Visual:

Ad Views $\rightarrow$ Clicks $\rightarrow$ Purchase



Expected Output:

Ad Views = 18

Clicks = 6

Prediction = Customer Will Purchase

Conclusion:

Bayesian Logistic Regression combines prior information with observed advertising behaviour to estimate purchase probability.

Q8. Linear Regression for Monthly Sales Forecasting

Algorithm Used:

Linear Regression

Model / Formula:

$$\text{Sales} = \beta_0 + \beta_1(\text{Advertising Spend}) + \beta_2(\text{Season Index})$$

Features:

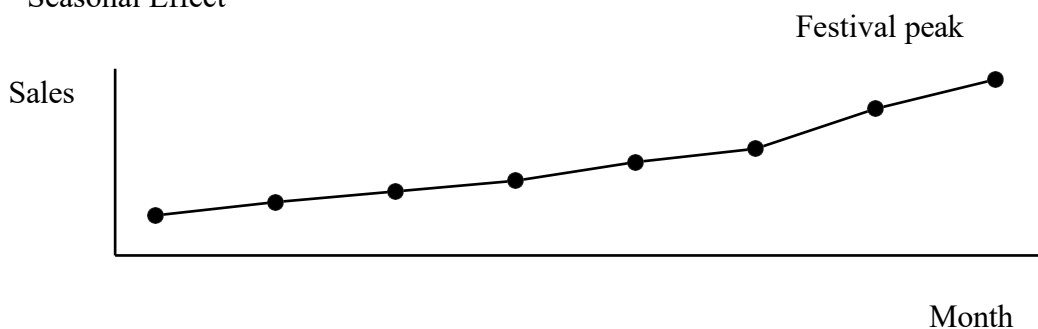
- Advertising Spend = Monthly promotional budget
- Season Index = Numerical representation of seasonal demand

Steps:

1. Collect previous monthly sales data.
2. Train the regression model.
3. Estimate the coefficients.
4. Enter the planned advertising expenditure and season index.
5. Forecast sales for the coming month.

Related Visual:

Monthly Sales Trend with
Seasonal Effect



Expected Output:

Advertising Spend =

₹2,50,000 Season =

Festival

Predicted Sales =

₹14,20,000

Conclusion:

Linear Regression can forecast continuous sales values by learning from advertising investment and seasonal patterns.

Q9. Naïve Bayes for News Category Classification

Algorithm Used:

Multinomial Naïve Bayes

Model / Formula:

$$P(\text{Category} | \text{Article}) \propto P(\text{Category}) P(\text{Words} | \text{Category})$$

Features:

- Sports
- Politics
- Entertainment

Steps:

1. Collect labelled news articles.
2. Extract important words or tokens.
3. Calculate category prior probabilities.
4. Estimate word likelihoods for each category.
5. Assign the article to the category with the highest posterior probability.

Related Visual:

Keyword-Based News Classification



Expected Output:

Article keywords = "Match", "Goal",
"Cricket" Prediction = Sports

Conclusion:

Multinomial Naïve Bayes classifies text efficiently by using word-frequency information and category probabilities.

Q10. Bayesian Logistic Regression for Student Placement Prediction

Algorithm Used:

Bayesian Logistic Regression

Model / Formula:

$P(\text{Placement}) = 1 / (1 + e^{-z})$, where $z = \beta_0 + \beta_1(\text{CGPA}) + \beta_2(\text{Coding}) + \beta_3(\text{Aptitude})$

Features:

- CGPA
- Coding Skill Level
- Aptitude Score

Steps:

1. Collect student academic and skill records.
2. Specify prior distributions for the coefficients.
3. Train the Bayesian logistic model.
4. Estimate placement probability for a new student.
5. Classify the student as Placed or Not Placed using the predicted probability.

Related Visual:

Placement Probability from Academic &
Skill Features



Expected

Output:

CGPA = 8.7

Coding Skills =

High Aptitude =

88 Prediction =

Placed

Conclusion:

Bayesian Logistic Regression estimates placement probability by combining academic performance, technical skills, and prior information.